

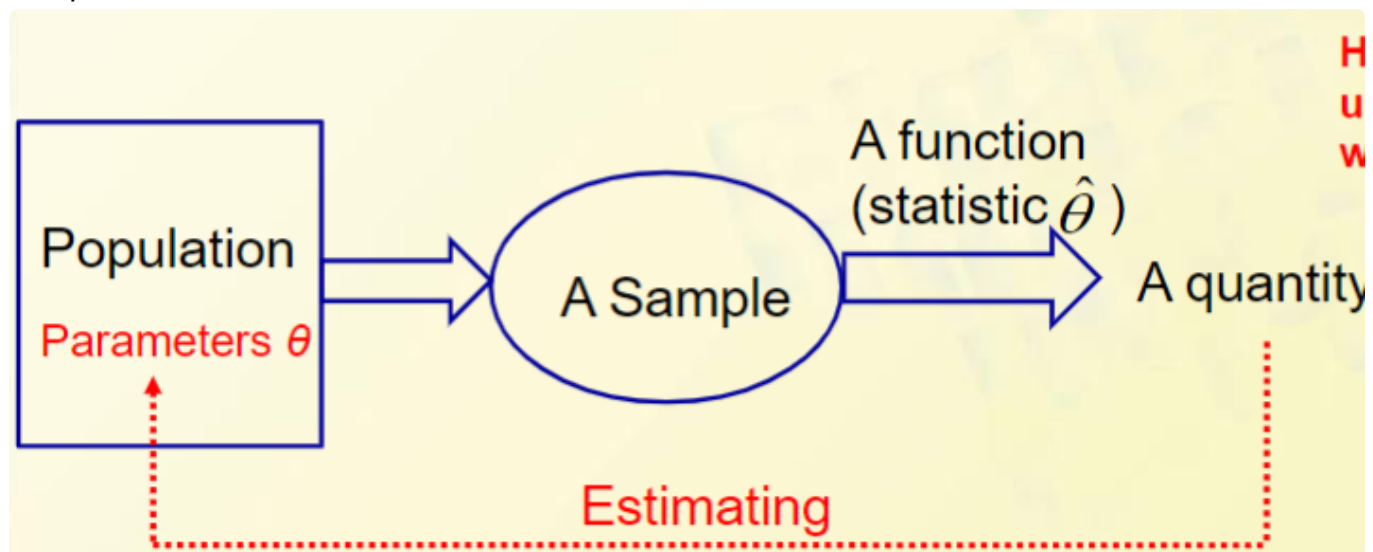
Chapter 6 Point Estimation

6.1 Some General Concepts of Point Estimation

Typically, we use θ as the parameter that describe the population characteristics.

The point estimate of a parameter θ is a **single number** that can be regarded as a **sensible value** for θ .

A point estimate is obtained by **selecting a suitable statistic and computing its value from the given sample data**. The selected statistic is called the **point estimate** of θ . Estimator, denoted by $\hat{\theta}$, is the random variable to estimate the parameter. Estimate is the concrete value from the sample.



The **type of population under study is usually known**, while the **parameters are unknown**.

Estimation Error Analysis

$$\hat{\theta} = \theta + \text{error of estimation}$$

A good estimator should have two properties:

- Unbiasedness(i.e., the **average error** should be zero)
- Minimum variance(i.e., the variance of error should be small)

Unbiased Estimator

A point estimator $\hat{\theta}$ is an **unbiased estimator** of θ if

$$E(\hat{\theta}) = \theta \text{ for every possible value of } \theta$$

If $\hat{\theta}$ is not unbiased, the difference $E(\hat{\theta}) - \theta$ is called the bias of $\hat{\theta}$.

Proposition

When X is a binomial rv with parameters n and p , the sample proportion $\hat{p} = X/n$ is an unbiased estimator of p .

Proposition

Let $X_1, \dots, X_n$ be a random sample from a distribution with mean μ and variance σ^2 , then the estimator

$$\hat{\sigma}^2 = S^2 = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

is an unbiased estimator of σ^2 , namely $E(S^2) = \sigma^2$. (note the denominator is $n - 1$)

Proposition

If $X_1, \dots, X_n$ is a random sample from a distribution with mean μ , then $\bar{X}$ is an unbiased estimator of μ . If in addition the distribution is continuous and symmetric, then the median and any trimmed mean are also unbiased estimator of μ .

Estimators with Minimum Variance

By the degree of centralizing. More centralized, less σ .

Estimator Selection

- When choosing among several different estimators of θ , select one that is unbiased
- For all unbiased θ , choose one that has minimum variance. (called minimum variance unbiased estimator (MVUE) of θ).

Theorem

Let $X_i, 1 < i < n$ be a random sample from a normal distribution= with μ and σ , then the estimator

$$\hat{\mu} = \bar{X}$$

is the MVUE for μ .

For μ , there exist four estimators: mean, median, trimmed mean, mean of extreme value

- For normal distribution, $\bar{X}$ is best of four estimators
- For Cauchy distribution, the median is best. Bad for $\bar{X}$ and $\bar{X}_e$.
- For uniform distribution, $\bar{X}_e$ is best.

- The trimmed mean is not best for all three situations, but it is a robust estimator.

The Standard Error

The standard error of an estimator $\hat{\theta}$ is its standard deviation $\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$

6.2 Methods of Point Estimation

Two constructive methods for obtaining point estimators:

- method of moments
- maximum likelihood estimation

The Method of Moment(矩)

The basic idea of this method is **equate certain sample characteristics** such as the mean **to the corresponding population** expected values.

Moment

Let $X_1, \dots, X_n$ be random sample from pmf or pdf. The k th population moment, or **k th moment of the distribution** $f(x)$, is

$$E(X^k)$$

The **k th sample moment** is

$$\frac{1}{n} \sum_{i=1}^n X_i^k$$

Moment Estimator

Using **sample moment replaces the population moment**, and find unknown parameter by equations.

■ Example 6.12

Let $X_1, X_2, \dots, X_n$ represent a random sample of service times of n customers at a certain facility, where the underlying distribution is assumed exponential with parameter λ . How to estimate λ by using the method of moments?

Step #1: The 1st population moment $E(X) = 1/\lambda$
then we have $\lambda = 1/E(X)$

Step #2: Use the 1st sample moment $\bar{X}$ to represent 1st population moment $E(X)$, and get the estimator

$$\hat{\lambda} = 1 / \bar{X}$$

■ Example 6.13

Let $X_1, \dots, X_n$ be a random sample from a gamma distribution with parameters α and β . Its pdf is

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

There are two parameter need to be estimated, thus, consider the first two moments

- $E(X) = \mu = \alpha\beta, E(X^2) = V(X) + [E(X)]^2 = \alpha\beta^2 + \alpha^2\beta^2$
- $\bar{X} \rightarrow E(X), \frac{1}{n} \sum X_i^2 \rightarrow E(X^2).$
and by two equations,

$$\hat{\alpha} = \frac{\bar{X}^2}{\frac{1}{n} \sum X_i^2 - \bar{X}^2}, \hat{\beta} = \frac{\frac{1}{n} \sum X_i^2 - \bar{X}^2}{\bar{X}}$$

Maximum Likelihood Estimation

The basic idea is that we look at the sample values and then **choose as our estimates** of the unknown parameters the values for which the pdf or pmf get maximum.

■ Example 6.15

A sample of ten new bike helmets manufactured by a certain company is obtained. Upon testing, it is found that the first, third, and tenth helmets are flawed, whereas the others are not. Let $p = P(\text{flawed helmet})$ and define $X_1, \dots, X_{10}$ by $X_i = 1$ if the i th helmet is flawed and zero otherwise. Then the observed x_i 's are 1,0,1,0,0,0,0,0,0,1.

The Joint pmf of the sample is

$$f(x_1, x_2, \dots, x_{10}) = p(1-p)p \cdots p = p^3(1-p)^7$$

**For what value of p is the observed sample most likely to have occurred?
Or, equivalently, what value of the parameter p should be taken so that the joint pmf of the sample is maximized?**

take logarithm and take derivative, get $p = 0.3$.

Three steps:

- write the joint pmf/pdf(Likelihood function):
 $f(x_1, \dots, x_n; \theta_1, \dots, \theta_m) = f(x_1; \theta_1, \dots, \theta_m) \cdots f(x_n; \theta_1, \dots, \theta_m)$ (if independent).
- Get the logarithm(if necessary)
- take the partial derivative with respect to θ_i and equals them to 0 and solve the resulting equations.